Unit-basis split table: evidence report

Date: 2026-10-19
Scope: the unit-basis decomposition of digits for the parent 17-position `140−τ` secp256k1 recoding, its two-bucket native table and the frozen Tier 1 split-versus-orbit batch comparison. This is a hot-table time/memory tradeoff, not a new scalar rule or an end-to-end ECDLP measurement.

Question and method

Can each selected digit `a+bτ` be represented in the unit basis `(1,1−τ)` with bounded coefficients, reducing retained table memory, and what throughput cost does this representation have against the parent's orbit table?

The exact screen checks all 19,183 selected digits and the terminal coefficient domain from the parent's selected 17-step trajectory. Lean proves the pair/scalar identity and the 92/90 coefficient bounds. The native overlay checks the parent Rust source hash before transforming it. Independent affine replay, exhaustive native residue checks and the 128-point native holdout compare output points against independent binary multiplication. The paired panel uses 128 frozen scalars, 96 passes per arm, alternating order and adjacent A/A controls; raw arms and host counters remain committed.

Curve and subgroup: secp256k1, `EC1P256Csecp256k1ha8dc9f115a15` in `crypto:docs/curves/registry.json`; $y^2=x^3+7 \bmod p$, prime subgroup order `n = FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD036`, cofactor 1, standard generator `G`. The experiment changes only the point-table representation; it adds no curve or isogeny.

Findings

- **Derived and checked:** `a+bτ=(a+b)−b(1−τ)` for every integer pair and integer `τ` eigenvalue. The selected digit components have magnitudes at most 92, and terminal components at most 90. Both buckets reuse the same per-position positive-multiple table; one unit-map rotation converts the second bucket before joining.
- **Independently replayed:** all 19,183 residues and 128 frozen native points passed; the independent affine replay matched binary multiplication for 388 scalars. The formal point-group/coordinate-action bridge remains open; implementation correctness is supported by these replays.
- **Measured, A/A-qualified, one RunPod allocation:** Tier 1 job `393dd4e822084bca846e4a6dd9192389`, instance `thyvmdqmzsz5ik` (RunPod `cpu3c`, EU-RO-1; 8 vCPU/16 GB). Captured host: x86_64 Ubuntu 24.04.4 under Docker, AMD EPYC 9655 96-Core Processor. The worker was pinned to CPU 13 from `{2,13,40,74,98,109,136,170}`; one NUMA node was reported. CPU quota was unlimited; host exclusivity and memory policy were unverified; physical memory type/speed/channel data remain unknown. All 96 arms and 48 pairs verified, artifacts remained stable, and all adjacent A/A controls passed [0.90,1.11]. The paired split/orbit throughput ratio was 0.736576 with repeat-bootstrap 95% interval [0.734478,0.738656] over 16 repeats on this allocation.
- **Resource tradeoff:** the native table retains 145,808 bytes versus 4,178,836 bytes for the orbit table; mean preparation was 90.61 versus 733.50 ms. The split form is 28.7 times smaller in retained bytes and prepares about 8.1 times faster in this run, while its online throughput was 0.7366 of the orbit reference. Setup is outside the hot-table timing. No one-call latency or full application speedup is claimed.

The plot shows the three candidate/reference ratios. Only throughput has a paired bootstrap interval; table bytes are exact retained-size counts and preparation values are reported means.

Visuals and graph audit

The editable unit-basis diagram and SVG show the two buckets and join. The plot source reads the committed analysis JSON and recreates the tradeoff plot. The PDF report embeds both figures.

Checked `docs/BENCHMARKS.md`, `docs/perf/PERFORMANCE_INDEX.md`, `docs/perf/weights.json`, and `experiments/volcano-ic/figures/` (`census.svg/.png`, `ecdlp.svg/.png`, `tau.png`). They cover library/suite kernels or IC/volcano results; this Rust fixed-base table-encoding comparison is separate. Its experiment-specific tradeoff graph is new; no canonical index row or existing graph is affected. The same curve remains registered in the sibling crypto registry; no local curve or route entry changes.

Open checks and reproduction

Fresh-allocation replication, one-call latency, and a formal point-group bridge remain separate follow-ups. No new benchmark was launched for this report. Reproduce the archived screen and Lean dependency build with commands in the README; the analysis script validates frozen receipts only.

Source records

- Experiment README and full accounting
- Analysis JSON, frozen panel manifest, panel rules
- Raw panel and manifests
- Screen, independent replay, Lean proof, and hash-pinned overlay
- Source revision before this report: PR #906 head `42283dcdc1ab5fbf1f7a8c51ffb857f51fad7022`

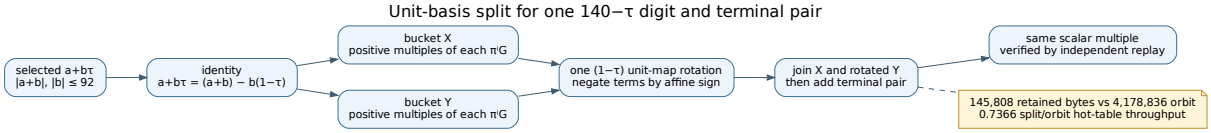
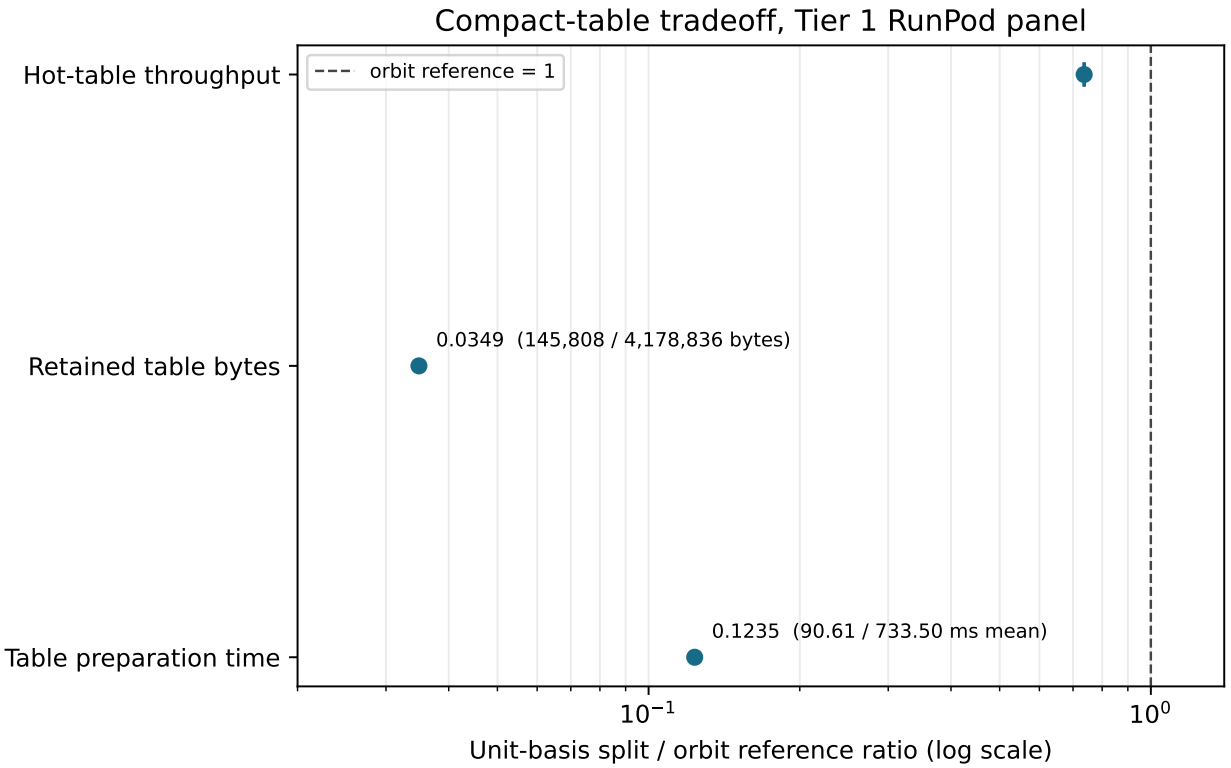


Figure 1: Unit-basis split and two point buckets



Throughput interval spans 16 repeats on one allocation; setup/memory ratios are reported separately.

Figure 2: Split/orbit throughput, retained bytes, and preparation ratios